Idea

🌍 Problem Statement

In India and globally, women often face unsafe situations, especially during:

- Late-night travel
- Isolated environments
- Public transport
- Emergencies where immediate help is unavailable

Key Issues:

- 🚫 No quick way to alert nearby help instantly
- 🚫 Panic situations prevent manual app usage
- 🚫 False alarms reduce trust in safety systems
- 🚫 Police response delay due to lack of real-time context

💡 Proposed Solution

Rakshika AI is a **smart emergency response system** that enables women to **trigger help instantly using natural gestures**, verifies the situation, and connects them to **nearby helpers + authorities in real time**.

⚙️ System Overview

The system consists of:

1. 📱 Mobile Application (Core System)

- Runs continuously in background
- Detects emergency triggers

- Performs validation
- Sends alerts

2. 🌐 Web Platform

- User registration & profile management
- Emergency contacts setup
- Admin & monitoring dashboard

🚀 Core Features & Workflow

1. ⚡ Instant Emergency Trigger

User can activate emergency mode using:

- Press **power button 3 times**
- Shake phone multiple times
- Voice trigger (e.g., “HELP”, “DANGER”)

👉 Designed for **zero interaction in panic situations**

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2. 🧠 Multi-Layer Legitimacy Check (Key Innovation)

To avoid false alarms, system performs validation:

Level 1: Manual Confirmation

- Quick popup with simple buttons
- Pattern-based tap input

Level 2: Physical Activity Check

- Phone shaking detection
- Motion sensor validation

Level 3: Voice Recognition

- Detect emergency words (“help”, “bachao”, etc.)

Level 4: AI-Based Context Verification (Fallback)

If user cannot respond:

- Capture **5-sec audio + video**
- AI analyzes:
  - Voice stress / panic tone
  - Background disturbance
  - Movement intensity

👉 Ensures **high accuracy + low false positives**

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3. 📍 Smart Emergency Alert System

Once verified:

Alerts sent to:

- 📌 Nearby registered users (community helpers)
- 👨‍👩‍👧 Emergency contacts
- 🚓 Nearby police stations

Includes:

- Live location tracking
- Audio/video snippet
- Emergency type

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4. 🤖 AI Emergency Caller / Chatbot

- Automatically calls emergency contacts
- Sends voice message:

> “User is in danger at this location”

- Chatbot communicates with helpers nearby

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5. 🧭 Nearby Helper Network (Unique Feature)

- Users can opt-in as **“Safety Volunteers”**
- Nearby users receive alerts and can respond

👉 Creates a **community-driven safety network**

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6. 🔄 Continuous Live Tracking

- Real-time GPS updates
- Route tracking
- Status monitoring until safe

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7. 🏥 Multi-Use Emergency Support

System can also be used for:

- Medical emergencies
- Theft situations
- Accidents

🧱 Technical Architecture

Frontend:

- Mobile App: Flutter / React Native
- Web: React.js

Backend:

- Node.js / Spring Boot
- REST APIs

Database:

- MongoDB / PostgreSQL

AI Components:

- Voice recognition model
- Basic emotion/stress detection
- Activity recognition

Integrations:

- Google Maps API (location tracking)
- Twilio / Firebase (calls & notifications)

🔐 Privacy & Safety

- Data encrypted
- Audio/video stored temporarily
- User consent for AI usage
- Secure authentication

🎯 Innovation Highlights (IMPORTANT FOR WINNING)

This is where you stand out:

- ✅ **Multi-layer verification system (very unique)**
- ✅ **AI used only as fallback (smart + practical)**
- ✅ **Community-based safety network**
- ✅ **Gesture-based activation (no UI dependency)**
- ✅ **Real-time evidence sharing (audio/video)**

📊 Impact

- 🚺 Enhances women safety in real time
- ⚡ Reduces emergency response time
- 🌐 Builds community safety ecosystem
- 🇮🇳 Scalable across India

📈 Future Scope

- Integration with government systems
- Smart wearable device support
- Predictive risk alerts using AI
- Public transport integration